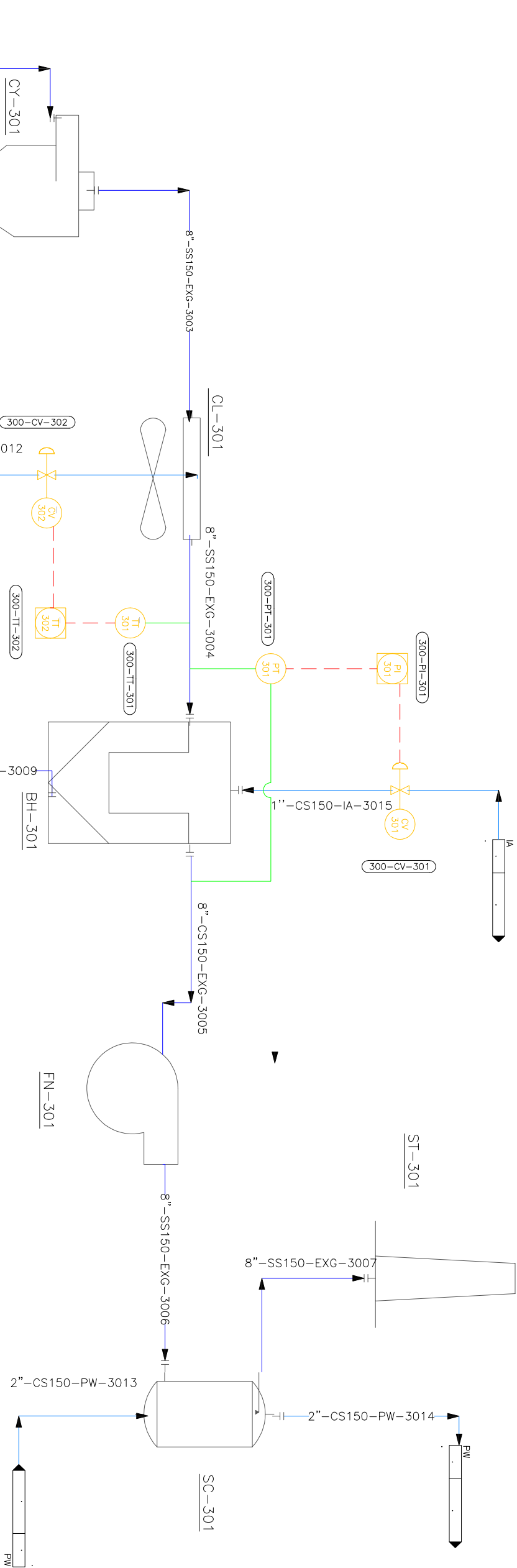




Tag Number	Category	Description
CV-301	Mechanical Equipment	Cyclone Separator
RL-302	Mechanical Equipment	Rotary Airlock Valve
CL-301	Mechanical Equipment	Evaporative Gas Cooler
BH-301	Mechanical Equipment	Baghouse Filter
FN-301	Mechanical Equipment	Rotary Airlock Valve
SC-301	Mechanical Equipment	Induced Draft (ID) Fan
ST-301	Mechanical Equipment	Wet Scrubber
TT-301	Instrumentation	Exhaust Stack
CV-302	Instrumentation	Temperature Transmitter
PT-301	Instrumentation	Control Valve
RL-301	Instrumentation	Pressure Transmitter
CV-301	Instrumentation	Pressure Indicator
8"-SS150-EXG-3001	Piping	Control Valve
8"-SS150-EXG-3002	Piping	Exhaust Gas Inlet Line
8"-SS150-EXG-3003	Piping	Exhaust Gas Line
4"-CS150-DST-3008	Piping	Cyclone Exhaust Gas Line
2"-CS150-PW-3012	Piping	Cyclone Dust Discharge Line
8"-SS150-EXG-3004	Piping	Quench Water Line
1"-CS150-IA-3015	Piping	Cooled Exhaust Gas Line
4"-CS150-DST-3009	Piping	Instrument Air Line
8"-SS150-EXG-3005	Piping	Baghouse Dust Discharge Line
8"-SS150-EXG-3006	Piping	Baghouse Exhaust Gas Line
2"-CS150-PW-3013	Piping	Fan Discharge Gas Line
2"-CS150-PW-3014	Piping	Scrubber Water Supply Line
8"-SS150-EXG-3007	Piping	Scrubber Effluent Line
8"-SS150-EXG-3007	Piping	Treated Exhaust Gas Line

General Notes

1. All pipes, fittings, and valves shall be designed in accordance with ASME B31.3 process piping standards.
2. All instrumentation symbols and nomenclature conform to ISA 5.1 standards.
3. Fail-safe actions for all pneumatic control valves
- (fail-open/fail-closed) shall be determined during the HAZOP review phase.
4. All exhaust ductwork (SS150) shall be thermally insulated to maintain off-gas temperature strictly above the acid dew point until discharge at ST-301.
5. Instrument air supply (ias) to baghouse (BH-301) pulse-jet cleaning system must be bone-dry (-40°C dew point) to prevent filter coking.
6. Wet scrubber (SC-301) effluent must be routed directly to the central wastewater treatment plant prior to discharge.

No.	Revision/Issue	Date

Drawing Name
Area300
Flue Gas Treatment and Heat Recovery

Project Name and Address
PAC_P&ID
Final Year Project
PMB, UPO, KNUST, Ghana
Kumasi Ashanti Region

Area 300	Stamp
Drawing Number 3	
Author GROUP 7B	